FOOD DELIVERY TIME PREDICTION

Final Project Report

Data Analysis and Machine Learning

Submitted as part of the practical assignment

1. Introduction

Food delivery time depends on different factors such as distance, traffic, weather, order priority and delivery person experience. In this project, the given food delivery dataset is analyzed and machine learning models are used to predict delivery time.

The project mainly covers data preprocessing, exploratory data analysis, feature engineering, Linear Regression and Logistic Regression.

2. Objective

• To understand the given food delivery dataset.

• To clean and prepare the data for machine learning.

• To study the relationship between different variables and delivery time.

• To predict delivery time using Linear Regression.

• To classify deliveries as Fast or Delayed using Logistic Regression.

• To evaluate the models using suitable performance measures.

3. Dataset

The dataset used in this project is Food_Delivery_Time_Prediction.csv. It contains 200 records and 15 columns. The columns include customer and restaurant locations, distance, weather, traffic, delivery person experience, order priority, order time, vehicle type, ratings, delivery time, order cost and tip amount.

4. Data Preprocessing

First, the dataset was checked for missing values and duplicate rows. There were no missing values and no duplicate rows, so no rows had to be removed or filled.

The categorical columns were converted into numerical form using one-hot encoding. The dataset was then divided into training and testing data using an 80:20 split. StandardScaler was used for the Logistic Regression model.

5. Exploratory Data Analysis

The average Delivery_Time is 70.495, while the median is 72.775. The minimum delivery time is 15.23 and the maximum is 119.67.

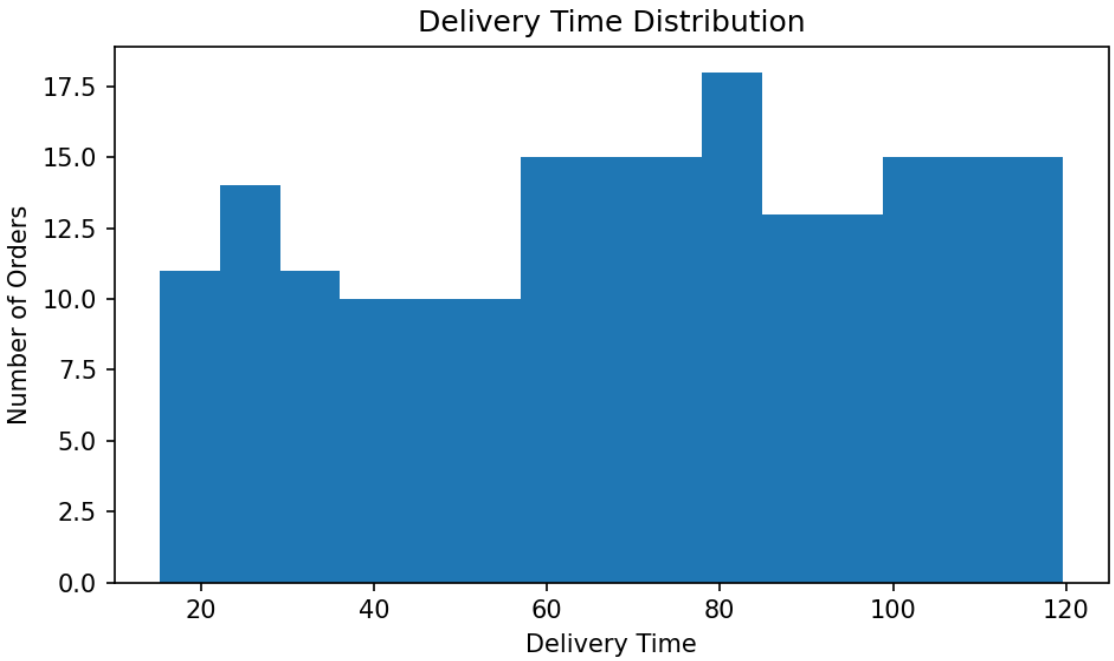


Figure 1: Distribution of delivery time.



Figure 2: Distance and delivery time.

5.1 Traffic Analysis

The average delivery time changes across traffic conditions. In the given data, the averages are 71.93 for Low traffic, 70.98 for Medium traffic and 67.57 for High traffic.

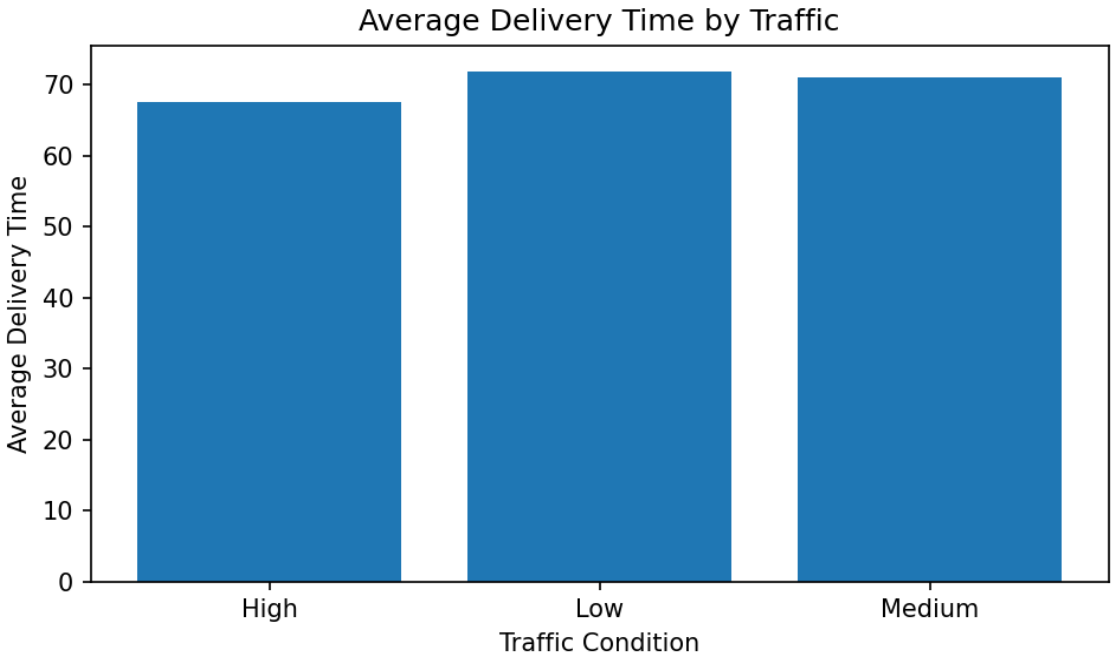


Figure 3: Average delivery time for different traffic conditions.

5.2 Correlation

Correlation was checked to see how the numerical variables are related to Delivery_Time. The relationships in this dataset are generally weak. Distance has a correlation of about -0.075 with Delivery_Time. The calculated geographic distance has a correlation of about -0.171.

6. Feature Engineering

The Haversine formula was used to calculate the distance between the customer and restaurant locations from their latitude and longitude values. A new Rush_Hour feature was also created. Morning and Evening were treated as rush hours, while Afternoon and Night were treated as non-rush hours.

7. Linear Regression

Linear Regression was used because Delivery_Time is a continuous value. The model was trained using the encoded features and the supplied Distance column. The test data was used to calculate MSE, MAE and R^2.

Measure	Result
Mean Squared Error (MSE)	1021.94
Mean Absolute Error (MAE)	27.18
R^2 Score	-0.105

The MAE is about 27.18. The R^2 value is negative, which shows that this model did not give a good fit for the test data.

8. Logistic Regression

For Logistic Regression, the median delivery time (72.775) was used to divide the orders into two groups. Orders at or below the median were marked Fast and orders above the median were marked Delayed.

The model was evaluated using accuracy, precision, recall, F1-score and the ROC curve.

Measure	Result
Accuracy	37.5%
Precision	0.368
Recall	0.350
F1-score	0.359
AUC	0.37

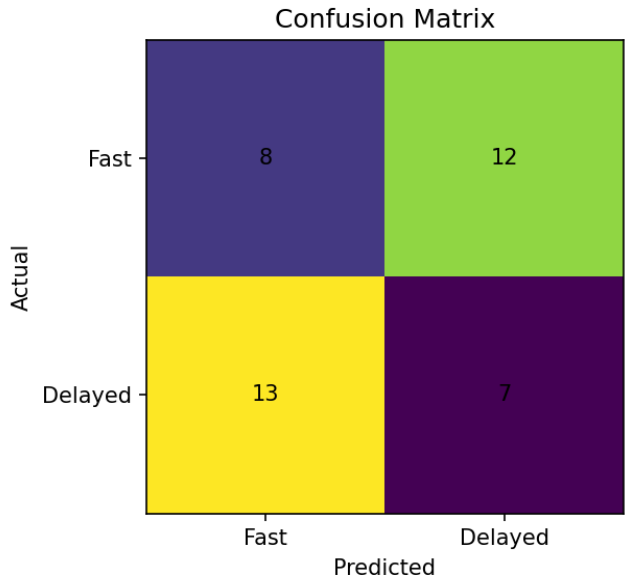


Figure 4: Confusion matrix.

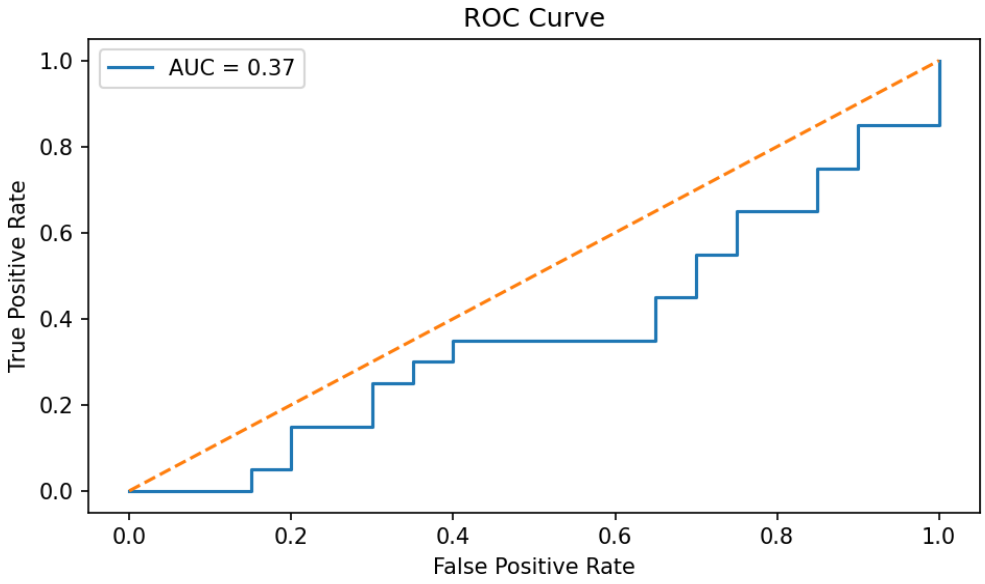


Figure 5: ROC curve.

9. Results and Discussion

The two models were used for different purposes. Linear Regression predicts the actual delivery time, while Logistic Regression predicts whether the delivery is Fast or Delayed.

The Linear Regression model gave an MSE of 1021.94, MAE of 27.18 and R^2 of -0.105. Logistic Regression gave 37.5% accuracy, 0.368 precision, 0.350 recall, 0.359 F1-score and 0.37 AUC.

The results are not very strong. This suggests that the variables available in the dataset do not explain delivery time very well. More information about the actual delivery process may be needed.

10. Possible Improvements

• Use a larger dataset with more delivery records.

• Include actual traffic information during delivery.

• Use route distance instead of only the supplied distance.

• Include restaurant preparation time.

• Include more detailed time information and delivery-person information.

• Try other machine learning models and compare their results.

11. Conclusion

In this project, the food delivery dataset was cleaned and analyzed using Python. Different graphs were used to understand the data. Feature engineering was also performed by calculating geographic distance and creating a rush-hour feature.

Linear Regression and Logistic Regression were then implemented. The obtained results show that the current models have limited predictive performance on the given dataset. However, the project demonstrates the complete process from data preprocessing and analysis to machine learning model evaluation.

12. References

• Food_Delivery_Time_Prediction.csv — dataset provided for the assignment.

• Food Delivery Prediction.ipynb — analysis and model implementation.

• Python Pandas and NumPy libraries.

• Matplotlib and Seaborn libraries.

• Scikit-learn library.